$\Rightarrow \quad (x-2)^2 = 5(y+1)$ which is in standard form. Therefore, $4a = 5$ and the vertex is $(2,-1)$, and the focus is $\left(2, \frac{1}{4}\right)$.

Equation of directrix is $y - k + a = 0$

$$y + 1 + \frac{5}{4} = 0$$

$$4y + 9 = 0 .$$

Length of latus rectum 5 units.

Example 5.20

Find the equation of the ellipse with foci $(\pm 2, 0)$, vertices $(\pm 3, 0)$.

Solution

From Fig. 5.36, we get

$SS' = 2c$ and $2c = 4$; $A'A = 2a = 6$

$\Rightarrow \quad c = 2$ and $a = 3$,

$\Rightarrow b^2 = a^2 - c^2 = 9 - 4 = 5$.

Major axis is along x-axis, since $a > b$.

Centre is (0, 0) and Foci are $(\pm 2, 0)$.

Therefore, equation of the ellipse is $\frac{x^2}{9} + \frac{y^2}{5} = 1$.

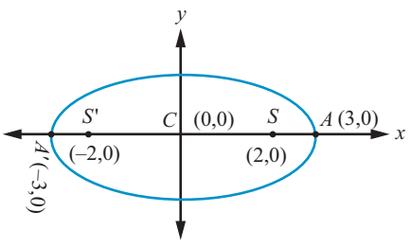


Fig.5.36

Example 5.21

Find the equation of the ellipse whose eccentricity is $\frac{1}{2}$, one of the foci is $(2,3)$ and a directrix is $x = 7$. Also find the length of the major and minor axes of the ellipse.

Solution

By the definition of a conic, $\frac{SP}{PM} = e$ or $SP^2 = e^2 PM^2$.

Then, $(x-2)^2 + (y-3)^2 = \frac{1}{4}(x-7)^2$

$\Rightarrow \quad 3x^2 + 4y^2 - 2x - 24y + 3 = 0$

$\Rightarrow \quad 3\left(x - \frac{1}{3}\right)^2 + 4(y-3)^2 = 3\left(\frac{1}{9}\right) + 4 \times 9 - 3 = \frac{100}{3}$

$\Rightarrow \quad \frac{\left(x - \frac{1}{3}\right)^2}{\frac{100}{9}} + \frac{(y-3)^2}{\frac{100}{12}} = 1$ which is in the standard form.

Therefore, the length of major axis $= 2a = 2\sqrt{\frac{100}{9}} = \frac{20}{3}$ and

the length of minor axis $= 2b = 2\sqrt{\frac{100}{12}} = \frac{10}{\sqrt{3}}$.

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